INTERNATIONAL MARITIME ORGANIZATION

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SUB-COMMITTEE ON NAVIGATION, COMMUNICATIONS AND SEARCH AND RESCUE
13th session
Agenda item 3

NCSR 13/3/5
20 March 2026
Original: ENGLISH
Pre-session public release: ☒

ROUTEING MEASURES AND SHIP REPORTING SYSTEMS

Amendments to the traffic separation scheme "Off New York"

Submitted by the United States

SUMMARY

Executive summary:	This document contains a proposal for an amendment to the traffic separation scheme *Off New York*. This proposed amendment is intended to maintain traditional shipping routes and continue to ensure navigation lanes remain free of fixed structures.
Strategic direction, if applicable:	7
Output:	7.22
Action to be taken:	Paragraph 28
Related documents:	*Ships' Routeing*, 2019 edition; SOLAS regulation V/10; resolution A.572(14), as amended; MSC.1/Circ.1060, as amended; NCSR 13/3/4, NCSR 13/3/6, NCSR 13/3/7 and NCSR 13/INF.7

Introduction

1 The United States proposes to amend the traffic separation scheme (TSS) *Off New York* originally adopted by IMO on 4 September 1987, implemented on 5 October 1987, and last amended in 2010. The TSS consists of five parts: precautionary areas; eastern approach off Nantucket; eastern approach off Ambrose Light; south-eastern approach; and southern approach. Each approach consists of two traffic lanes and a separation zone: geographical coordinates and a chartlet depicting the TSS is provided in IMO's *Ships' Routeing*, edition 2019.

Summary

2 This document proposes to establish two new precautionary areas at the offshore termini of the TSS *Off New York*: south-eastern approach and southern approach. The proposed precautionary area at the terminus of the south-eastern approach would be an approximately 197-square nautical mile area. The new precautionary area at the terminus of the southern approach would be an approximately 146-square nautical mile area. In order to facilitate the establishment of these precautionary areas, approaches will need to be modified by shortening the length of the traffic lanes and separation zones. Figure 1 provides a side-by-side comparison of the current TSS with the proposed amendments.

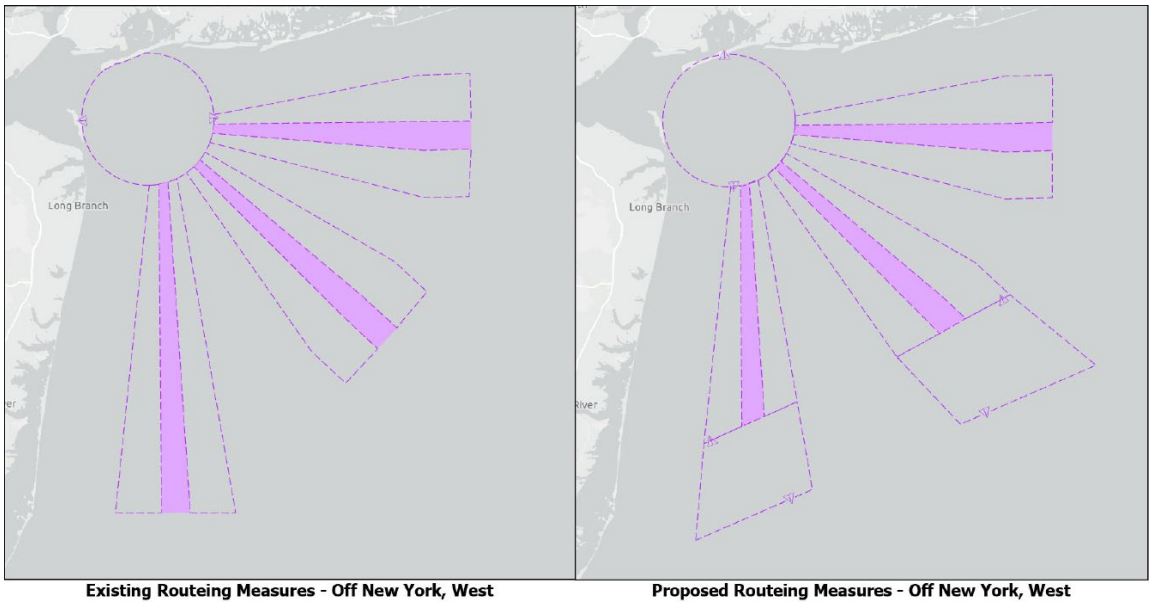

Figure 1: Proposed amendments to the TSS Off New York

3 The United States has placed particular attention on expanded oil, natural gas, hydrokinetic, solar, nuclear, aquaculture and other non-mineral energy resources on the outer continental shelf (OCS). In identifying viable locations in the United States exclusive economic zone (EEZ) for energy production, the United States has also identified safe access routes, or "fairways", for the movement of vessel traffic proceeding to or from United States ports, in which no artificial island or fixed structure, whether temporary or permanent, will be permitted. As such, the designation of a fairway recognizes the paramount right of navigation over all other uses in the designated areas. Based on the historical shipping trends provided in paragraph 9, coupled with the planned construction of offshore structures, the United States expects shipping transits in the approaches to New York to consolidate into these designated fairways. Figure 2 provides an overview of the proposal with United States fairways depicted in green and offshore energy construction sites in blue. A traditional north-east/south-west shipping route, as seen with AIS vessel movements in figure 3, converges perpendicular to the TSS *Off New York* at the offshore terminus of the southern and south-eastern approaches. By establishing a new precautionary area at the offshore terminus of these approaches, this proposal addresses the meeting of vessel traffic at a critical intersection, requiring mariners to navigate with particular caution in the designated areas. A precautionary area is the preferred routeing measure as it provides a charted identifiable area where ships must navigate with particular caution. A direction of traffic flow was considered, but determined to be unnecessary and inconsistent with all other precautionary areas throughout the United States. The amendment will help provide predictable navigation corridors to accommodate traditional shipping routes along with the evolving offshore energy infrastructure, thereby improving operational safety in increasingly congested areas of New York.

4 In addition, this document presents all geographical coordinates in the IMO *Ships' Routeing* 2019 edition description of the TSS *Off New York* converted from the existing North American 1927 geodetic datum to the World Geodetic System of 1984 datum. Updated coordinates that have changed with respect to the shift in datum are provided in the annex.

5 As noted in the annex, precautionary area (b) is located between part II of this traffic separation scheme (eastern approach, off Nantucket) and the traffic separation scheme *In the approach to Boston, Massachusetts*. Details of the precautionary area are contained in paragraph (b) of the description of the precautionary areas in the amended traffic separation scheme *In the approach to Boston, Massachusetts*. The TSS *In the approach to Boston, Massachusetts* is not being proposed for an amendment with this proposal.

6 A study, Port Access Route Study: Off New York, is referenced in document NCSR 13/INF.7. The study aimed to characterize whether amendments to existing, or additional routeing measures are necessary to improve navigation safety due to factors such as planned or potential offshore energy development, current port capabilities and planned improvements, increased vessel traffic, existing and potential anchorage areas, changing vessel traffic patterns, meteorological and oceanographic effects, or navigational difficulty in the proposal area. An in-depth analysis of the history of groundings, collisions, or damage to the marine environment can be found in section IV of the study. There has been no significant growth in these incidents that is precipitating the establishment of these precautionary areas. This proposal is submitted in advance of offshore construction as a preventative measure to ensure the safety of navigation in the area.

Description of the area

7 The locations of the proposed precautionary areas, existing TSS, United States fairways, and offshore energy areas are provided in figure 2. Geographic positions, number, edition and geodetic datum of the reference chart used to define the amended routeing system are set out in the annex.

Cooperation between interested States

8 The United States has all rights, jurisdiction and obligations to the waters in the proposal to amend the TSS Off New York. The TSS is entirely in the EEZ of the United States and there is no need for a cooperation agreement with other States.

Traffic considerations

9 The United States analysed vessel traffic in the waters off New York in the United States and focused on waters located seaward of existing port approaches within the United States EEZ. The study referenced in document NCSR 13/INF.7 allowed the United States to consider vessel movements among both domestic and international ports of call to inform marine planning in the area off New York. Information used included automatic identification system (AIS) data and information from shipping organizations to identify traditional navigation routes. The AIS data identified customary routes transited by vessels. Because vessel types have different manoeuvring capabilities and navigation safety requirements, the identification of their customary routes and required shipping route widths were studied and aggregated separately for evaluation. Although the most common vessel traffic in the area is entering/departing New York through the existing TSS as seen in figure 3, figure 4 shows consolidated AIS tracks for the northeasterly and southwesterly shipping traffic off New York. The traffic in red shows the highest concentration of shipping traffic in this direction, leading to the United States' concern for a significant meeting of perpendicular vessel traffic at the offshore terminus of the existing TSS.

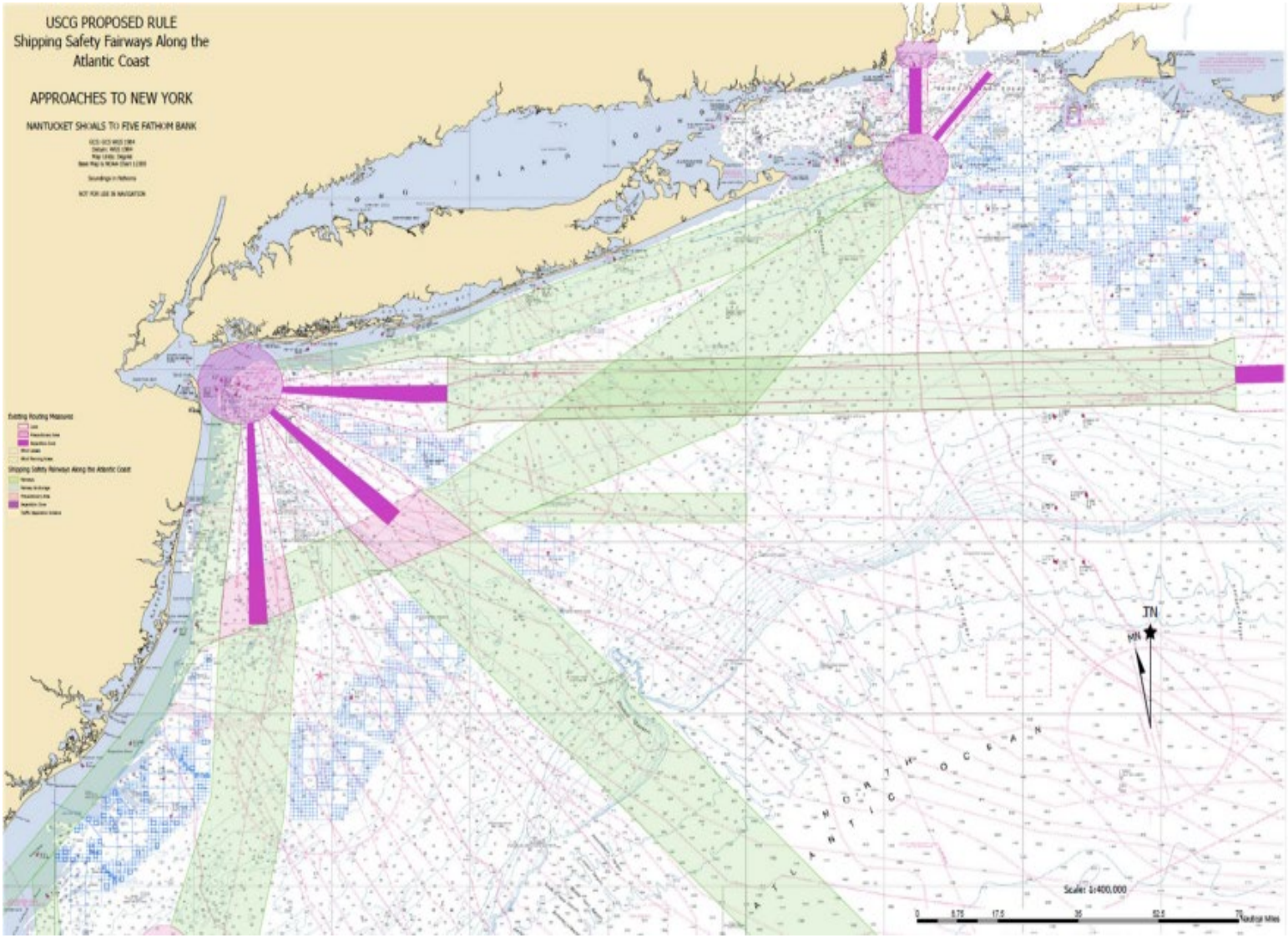

Figure 2: Proposed amendments with existing TSS, US fairways, and offshore energy areas

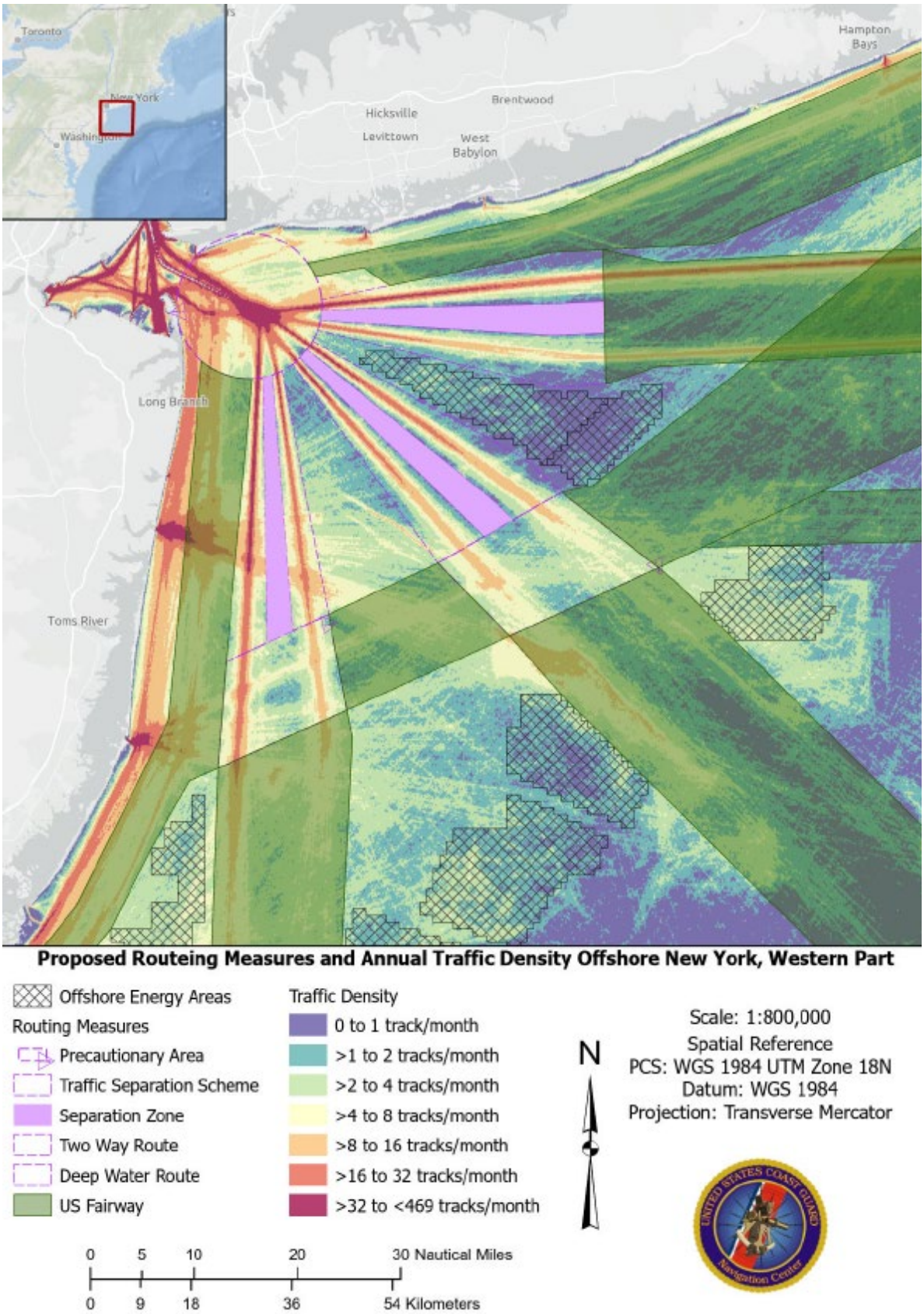

Figure 3: Total AIS vessel movements off New York

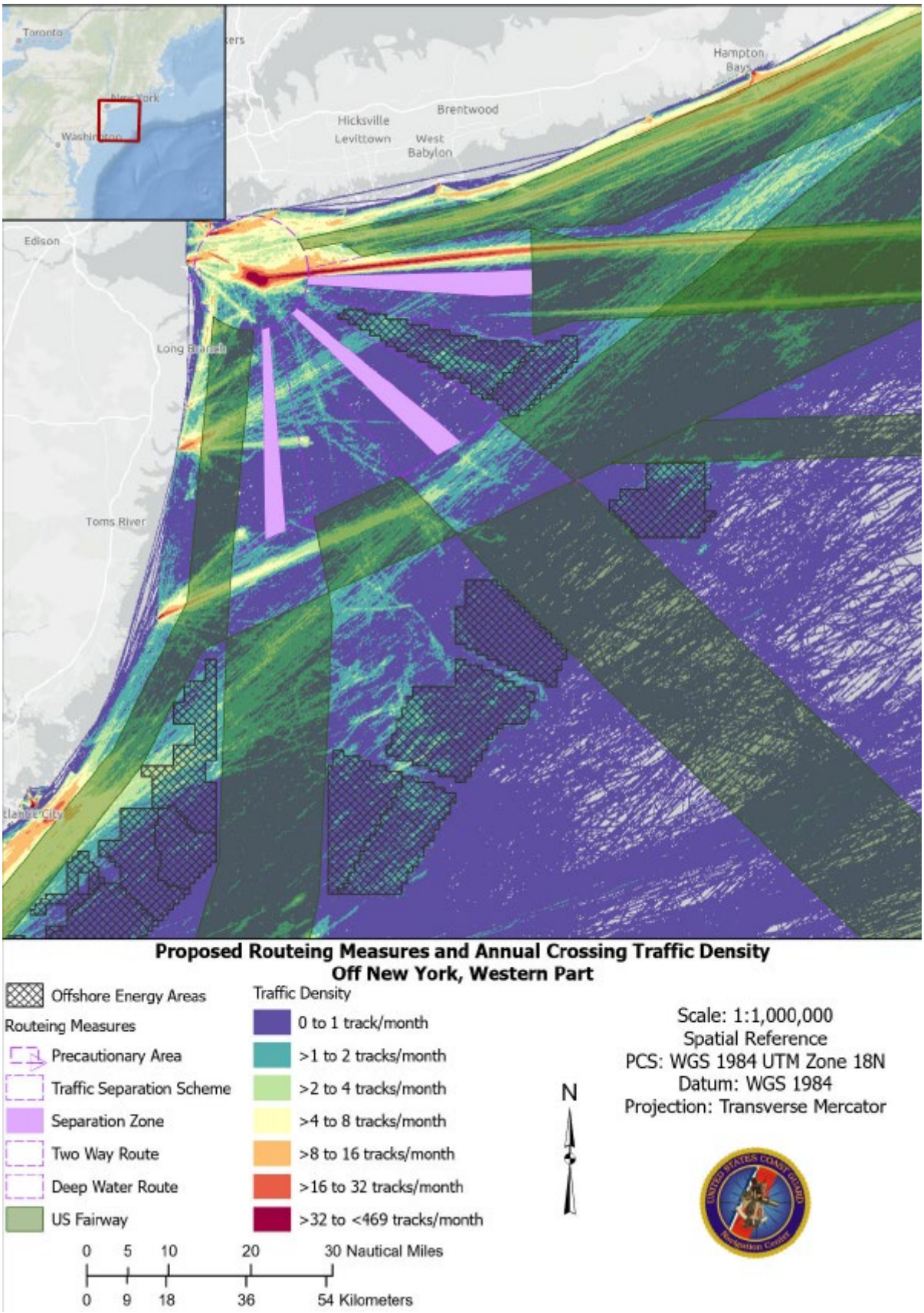

Figure 4: NE/SW traffic off New York

10 In addition to transiting perpendicular to the established TSS, the historical tracks cut through multiple offshore construction sites. The establishment of United States fairways, as denoted in figures 2, 3 and 4, will consolidate all shipping in the area and redirect northeast/southwest traffic to the north of the southwestern-most construction site and across the TSS through the proposed precautionary areas. Thus, safely navigating around offshore structures.

11 Aids to navigation (AtoN) in the vicinity of New York can be seen in figure 5. There are no AtoN marking the existing TSS. The entrance channel into New York Harbor is marked by Ambrose Channel Lighted Whistle Buoy A, which is located in the centre of the existing precautionary area at the inshore terminus of the three approaches. The entrance channel is further marked by a series of 10 lateral aids extending to the northwest from Ambrose Channel Lighted Buoy A. The Sandy Hook Channel is located to the south of the Ambrose Channel and is marked by one safe water mark, Scotland Lighted Buoy S, and 11 lateral buoys.

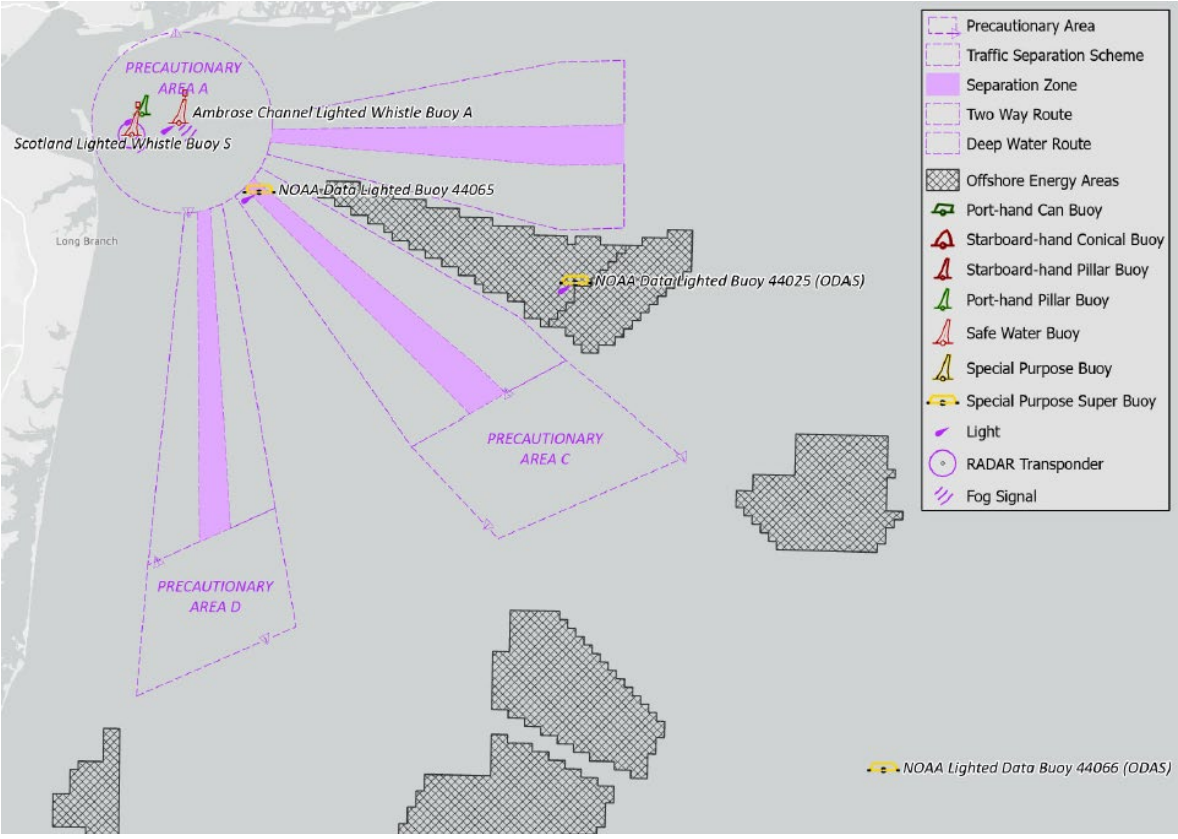

Figure 5: Aids to navigation off New York

12 Several private meteorological and oceanographic buoys are located between the eastern approach, off Ambrose Light and the south-eastern approach. These are buoys for research purposes associated with an ongoing offshore energy construction project and not for navigational use.

13 The United States has series of red lateral buoys marking a coastal along the Atlantic seacoast (not seen). Barnegat Offshore Lighted Buoy 2, red starboard-hand buoy is part of this marking system and is not associated with these routeing measures. The buoy's location, however, marks safe passage past a wreck and is in the vicinity of the turn for the new northeast/southwest United States fairway. The United States does not intend to move this buoy upon establishment of the proposed precautionary areas, as it is not associated with the existing TSS and subsequent routeing measures.

14 Physical AtoN were considered in the vicinity of the proposed precautionary areas; however, given the nature of the vessel traffic in the area, the United States determined that physical AtoN might obstruct the flow of safe navigation through the precautionary areas. Virtual AtoN were also considered for the precautionary areas, but while the current transmission capability could provide for virtual AIS AtoN in the proposed precautionary area at the offshore terminus of the southern approach, the existing infrastructure could not support a virtual AIS AtoN in the proposed precautionary area at the offshore terminus of the south-eastern approach. This coupled with the fact that no AtoN currently mark the existing TSS, led the United States to determine that provide a single virtual AIS AtoN in a precautionary area at the offshore terminus of the southern approach is unnecessary and would be inconsistent with the current marking scheme. Further use of AIS AtoN will be considered in the future as the capability improves. Consequently, AtoN in the area do not need to be amended or supplemented.

15 Precautionary area A, as provided in the annex, is described as "centred upon Ambrose Light". Ambrose Light and the corresponding structure were discontinued in 2008 and permanently replaced with Ambrose Channel Lighted Whistle Buoy A. Thus, the United States proposes to remove reference to Ambrose Light in the description of precautionary area A. The referenced position remains unchanged. This position has been updated in the annex and labelled position (44) on the enclosed chartlet.

16 A vessel traffic service (VTS) is provided for entry into New York; however, the VTS does not extend beyond Ambrose Channel Lighted Buoy A and is therefore not available in the vicinity of the proposed precautionary areas.

17 The United States National Oceanic and Atmospheric Administration (NOAA) hydrographic surveys for US3NY01M edition 2024 are based on International Hydrographic Organization (IHO) standards, including special publication 44. These surveys accurately represent the water depths in the area, and the depths provided on the nautical charts. All paper charts in the United States were discontinued by January 2025. Corresponding ENCs are provided and updated in the annex as appropriate.

18 Eight offshore energy areas are located in the general vicinity of the proposal as seen in figure 2. United States law prohibits the construction of structures within an established TSS, precautionary area or designated fairway.

Marine environmental considerations

19 The prevailing wind directions in the area are along the coast during the summer months and across the coast during the winter months. Average wind speed is 14-18 knots with seas of 2-3 feet. The strongest winds are observed during the presence of a hurricane off the coast of the United States Atlantic; the easterly cyclonic winds along the northern edge of the storm could drive a strong surge to the west, laterally along the southern coast of Long Island. A few times per year, primarily in the winter months, the wind direction is north-east to north.

20 The United States considered vessel noise, vessel operations and vessel movements in the area of the proposal. These considerations are not expected to result in significant impacts to the following resources: air quality, ambient sound, marine vegetation, marine invertebrates, flying insects, birds, bats, marine fish, essential fish habitat, commercial fishing, marine construction, mineral extraction, oil and gas extraction, recreation and tourism, existing renewable energy projects, research, transportation and shipping, and subsistence fishing and hunting.

21 There are no marine environmental considerations that will impact this proposal.

22 Additional information regarding marine and environmental considerations can be found in document NCSR 13/INF.7.

Status of the routeing system

23 This proposal is recommendatory and is designed to preserve traditional maritime routes and safe access to United States ports and protect them from fixed structures that could impact navigation safety and generally prioritize the right of navigation over all other uses.

Position-fixing in relation to the routeing system

24 Accurate position-fixing is widely available to all vessel traffic using the global positioning system (GPS). Aids to navigation, including the establishment of virtual AtoN, are addressed in paragraphs 11 through 15. Because of low user numbers and significant operational costs compared to satellite-based GPS, LORAN-C was considered unnecessary for modern navigation and discontinued in February 2010. Nautical charts for the area are available. GPS and nautical charts used in conjunction with AIS provide real-time location data to other vessels and shore stations.

Miscellaneous information

25 Prior to establishing the fairways, as discussed in paragraph 3, the United States conducted a Port Access Route Study (PARS) to determine traffic density; if current vessel routeing measures are adequate or require modifications; the type of new vessel routeing measures as necessary; and if the usage of the vessel routeing measures should be mandatory for specific classes of vessels. As provided in paragraph 6, a copy of this study is provided in document NCSR 13/INF.7.

26 Mariner participation in a PARS is essential for the United States to gather first-hand operational data, ensuring that proposed, updated, or new fairways and routeing measures safely accommodate vessel traffic while mitigating risks from offshore development and congestion. Local pilots' associations, local port authorities, commercial fishing, passenger vessel operators, bulk and container vessel operators, and recreational boaters were consulted throughout the study to capture direct input regarding the proposed area. Pertinent information has been provided in this proposal. Additional information captured from the consultations and subsequent analysis is provided in document NCSR 13/INF.7.

Proposed date of implementation

27 It is proposed that the amendment enter into force six months after adoption by the Committee.

Action requested of the Sub-Committee

28 The Sub-Committee is invited to consider the proposal in the annex and recommend it to the Committee for adoption.

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ANNEX

PROPOSED AMENDED TRAFFIC SEPARATION SCHEME AND ASSOCIATED MEASURES "OFF NEW YORK"

The highlighted areas are the proposed changes.

OFF NEW YORK

(Reference charts: United States ~~12300, 1985 edition; 12326, 1986 edition~~ US3NY01M, 2024 edition.

Note: ~~These charts are~~ This chart is based on ~~North American 1927 geodetic~~ World Geodetic System of 1984 datum.)

Description of the traffic separation scheme

The traffic separation scheme *Off New York* consists of five parts:

Part I

Precautionary areas

(a) ~~A p~~Precautionary area A with a radius of seven miles is centred upon ~~Ambrose Light in~~ geographical position (44) 40°27′.50 N, 073°49′.87 W.

(b) ~~A p~~Precautionary area B is located between part II of this traffic separation scheme (eastern approach, off Nantucket) and the traffic separation scheme "In the Approach to Boston, Massachusetts". Details of the precautionary area are contained in paragraph (b) of the description of the precautionary areas in the amended traffic separation scheme "In the approach to Boston, Massachusetts".

(c) Precautionary area C is located at the offshore terminus of the south-eastern approach bounded by a line connecting the following geographical positions:

(27) 40°09′.02 N, 073°10′.82 W
(31) 40°02′.41 N, 073°26′.55 W
(32) 39°55′.23 N, 073°17′.71 W
(33) 40°01′.53 N, 072°58′.88 W

(d) Precautionary area D is located at the offshore terminus of the southern approach bounded by a line connecting the following geographical positions:

(38) 39°57′.63 N, 073°40′.41 W
(41) 39°53′.17 N, 073°53′.35 W
(42) 39°42′.92 N, 073°54′.53 W
(43) 39°48′.35 N, 073°38′.28 W

Part II

Eastern approach, off Nantucket

(a) A separation zone is bounded by a line connecting the following geographical positions:

(1) 40°28′.~~75~~76 N, 069°14′.~~83~~80 W
(2) 40°27′.~~62~~63 N, 070°13′.~~77~~74 W
(3) 40°30′.~~62~~63 N, 070°~~14′.00~~13′.97 W
(4) 40°31′.~~75~~76 N, 069°14′.~~97~~94 W

(b) A traffic lane for westbound traffic is established between the separation zone and a line connecting the following geographical positions:

(5) 40°36′.~~75~~76 N, 069°15′.~~17~~13 W (6) 40°35′.~~62~~63 N, 070°14′.~~15~~12 W

(c) A traffic lane for eastbound traffic is established between the separation zone and a line connecting the following geographical positions:

(7) 40°22′.~~62~~63 N, 070°13′.57 W (8) 40°23′.75 N, 69°13′.94 W

Part III

Eastern approach, off Ambrose Light

(a) A separation zone is bounded by a line connecting the following geographical positions:

(9) 40°24′.33 N, 073°04′.~~97~~94 W
(10) 40°24′.20 N, 073°11′.~~50~~47 W
(11) 40°26′.~~00~~01 N, 073°40′.~~93~~89 W
(12) 40°27′.~~00~~01 N, 073°40′.~~75~~70 W
(13) 40°27′.20 N, 073°11′.~~50~~47 W
(14) 40°27′.33 N, 073°04′.~~95~~92 W

(b) A traffic lane for westbound traffic is established between the separation zone and a line connecting the following geographical positions:

(15) 40°32′.33 N, 073°04′.~~95~~92 W
(16) 40°32′.20 N, 073°11′.~~50~~47 W
(17) 40°28′.00′ N, 073°40.~~73~~70′ W

(c) A traffic lane for eastbound traffic is established between the separation zone and a line connecting the following geographical positions:

(18) 40°25′.05 N, 073°41′.~~32~~26 W
(19) 40°19′.20 N, 073°11′.~~50~~47 W
(20) 40°19′.33 N, 073°04′.~~97~~94 W

Part IV

South-eastern approach

(a) A separation zone is established bounded by a line connecting the following geographical positions:

(21) 40°~~03′.10~~04′.95 N,073°~~17′.93~~20′.48 W
(22) 40°06′.~~50~~51 N,073°22′.~~73~~70 W
(23) 40°22′.~~45~~44 N,073°43′.~~55~~50 W
(24) 40°23′.~~20~~18 N, 073°42′.~~70~~63 W
(25) 40°08′.~~72~~73 N, 073°20′.~~10~~07 W
(26)40°~~05′.32~~06′.49N,073°~~15′.28~~16′.98 W

(b) A traffic lane for north-westbound traffic is established between the separation zone and a line connecting the following geographical positions:

(27) 40°~~08'.98~~09′.02 N, 073°10′.~~87~~82 W (29) 40°24′.~~02~~01 N, 073°41′.~~97~~90 W
(28) 40°12′.~~42~~43 N, 073°15′.~~67~~64 W

(c) A traffic lane for south-eastbound traffic is established between the separation zone and a line connecting the following geographical positions:

(30) 40°21′.~~82~~80 N, 073°44′.~~55~~50 W ~~(32) 39°59′.43 N, 073°22′.35 W~~
(31) 40°02′.~~80~~41 N, 073°~~27'.15~~26′.55 W

Part V

Southern approach

(a) A separation zone is bounded by a line connecting the following geographical positions:

(~~33~~34) 39°~~45'.70~~55′.01 N, 073°48′.~~00~~09 W (~~35~~36) 40°20′.~~87~~82 N, 073°47′.~~07~~04 W
(~~34~~35) 40°20′.~~63~~58 N, 073°48′.~~33~~30 W (~~36~~37) 39°~~45'.70~~56′.05 N,073°44′.~~00~~96 W

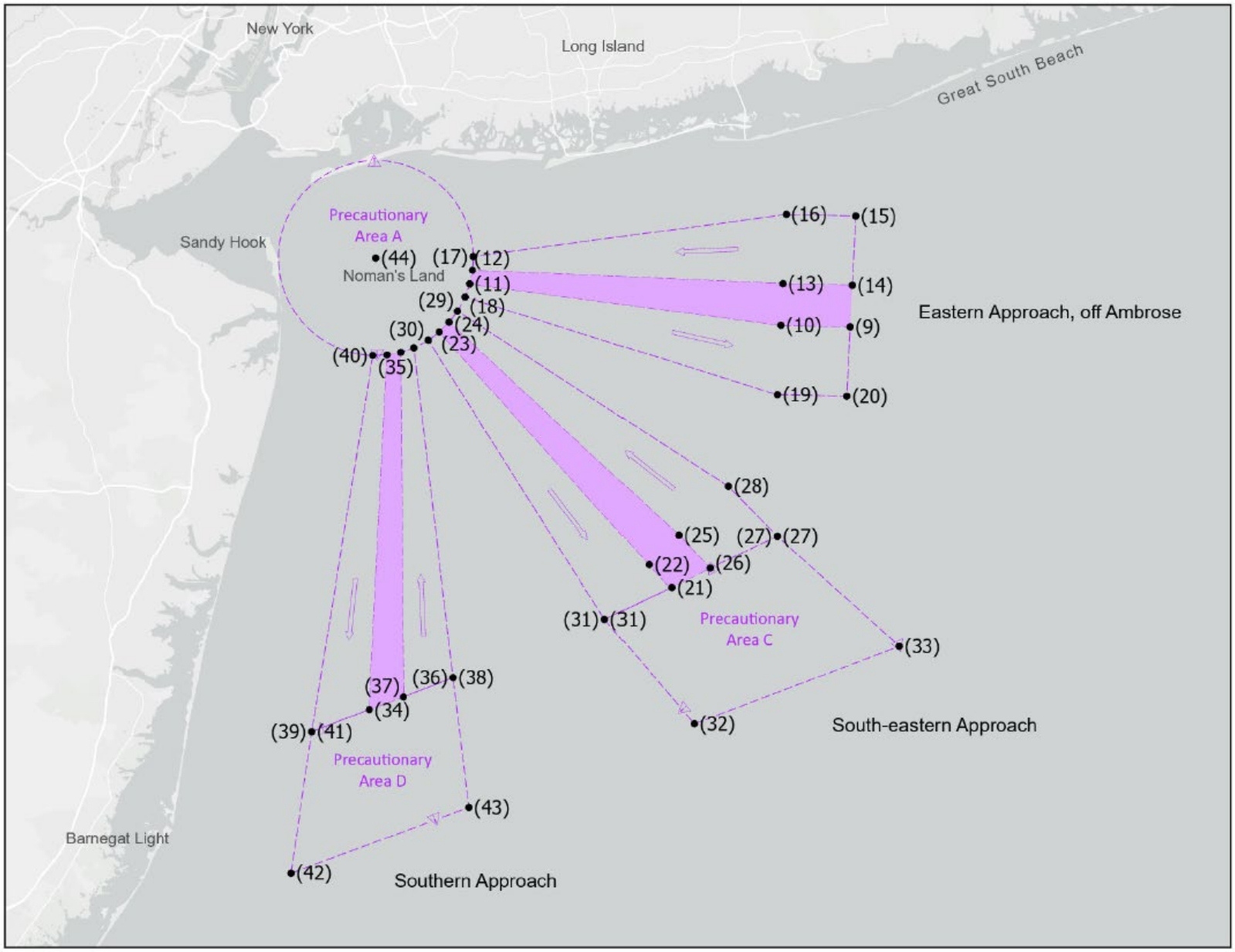
(b) A traffic lane for northbound traffic is established between the separation zone and a line connecting the following geographical positions:

(~~37~~38) 39°~~45'.70~~57′.63 N, 073°~~37'.70~~40′.41 W (~~38~~39) 40°21′.~~25~~20 N, 073°45′.~~85~~81 W

(c) A traffic lane for southbound traffic is established between the separation zone and a line connecting the following geographical positions:

(~~39~~40) 40°20′.~~53~~48 N, 073°49′.~~65~~63 W (~~40~~41)39°~~45'.70~~53′.17N,073°~~54'.40~~53′.35 W

~~**Note:** Use of LORAN-C enables masters of appropriately equipped ships to be informed highly accurately and continuously about the ship's position in the area covered by this scheme.~~


New York
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Great South Beach
Sandy Hook
Precautionary
Area A
Noman's Land
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South-eastern Approach
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Precautionary
Area D
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Barnegat Light
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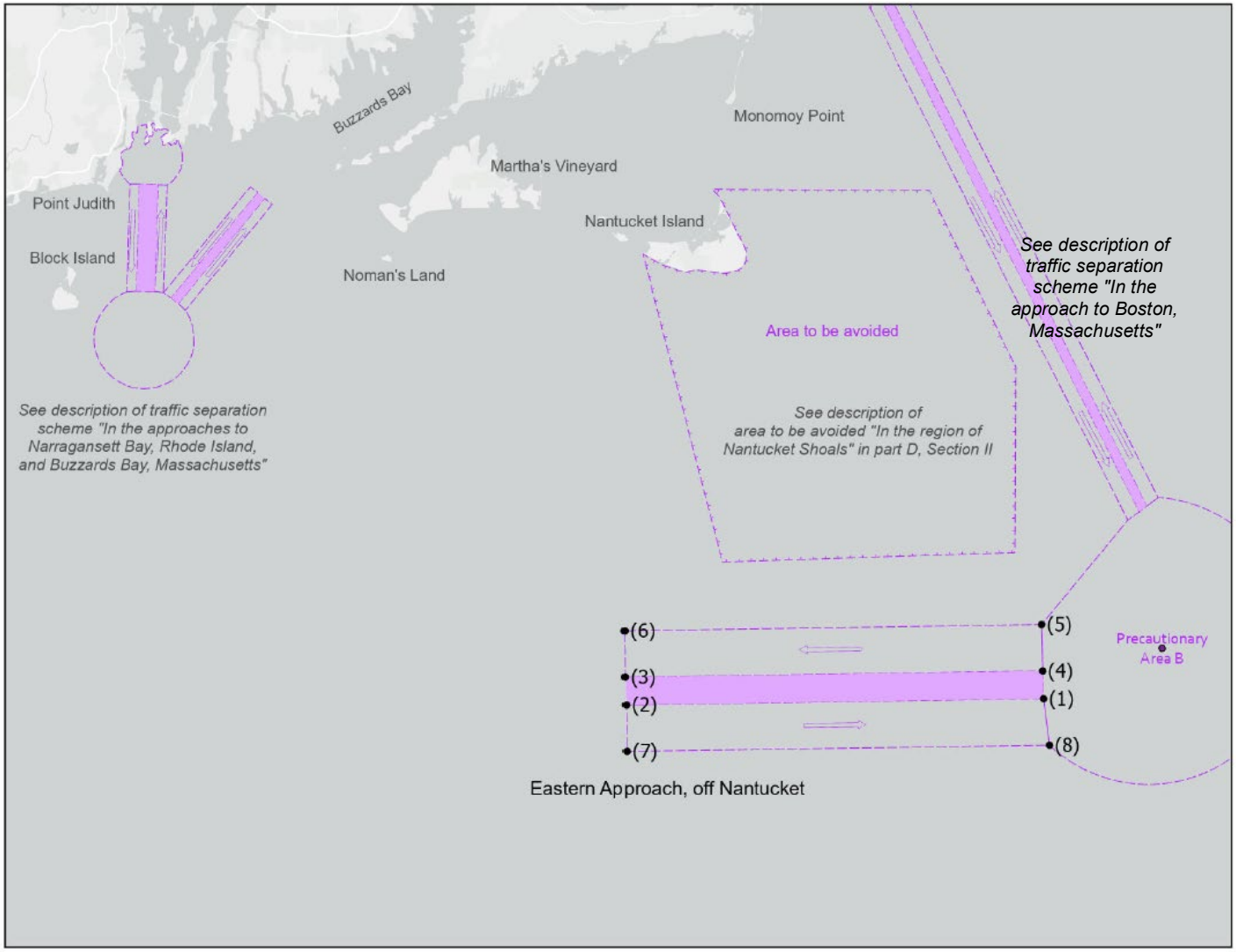
Southern Approach


Buzzards Bay
Monomoy Point
Martha's Vineyard
Point Judith
Nantucket Island
Block Island
Noman's Land
See description of traffic separation scheme "In the approach to Boston, Massachusetts"
Area to be avoided
See description of traffic separation scheme "In the approaches to Narragansett Bay, Rhode Island, and Buzzards Bay, Massachusetts"
See description of area to be avoided "In the region of Nantucket Shoals" in part D, Section II
(6)
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Precautionary Area B
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Eastern Approach, off Nantucket